CO-1
Assessment Tool – 2
ITA0329 – Mobile Computing

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1.
Given:
Total bandwidth = 45 MHz = 45000 kHz
Simplex channel bandwidth = 25 kHz

Formula:
No. of channels = Total bandwidth / (2 × channel bandwidth)

= 45000 / (2 × 25)
= 45000 / 50
= 900
Ans: 900

2.
Given:
Total channels = 560
Cluster size = 7
Formula: Channels per cell = Total channels / Cluster size

= 560 / 7
= 80
Ans:

3.
Given:
No. of carriers = 100
Time slots per carrier = 8
Formula: Total channels = Carriers × Time slots

= 100 × 8
= 800
Ans:

4.
Given:
Base station power = 80 W
Handset power = 4 W
Formula: Power ratio = $10 \log10(P_1/P_2)$

= 10 log10(80/4)
= 10 log10(20)

= 13.01 dB
Ans: 13.01 dB.

5.
Given:
Carrier power = -60 dBm
Interference power = -92 dBm
Formula: C/I = Carrier power - Interference power

= -60 - (-92)
= -60 + 92
= 32 dB
Ans: 32 dB.

6.
Given:
Total bandwidth = 42 MHz = 42000 kHz
Channel bandwidth = 300 kHz
Cluster size = 7
Total channels = 42000 / 300
= 140
Channels per cell = 140 / 7
= 20
Ans: 20

7.
Given:
Initial power = 64 W
Distance is doubled
Path-loss exponent = 4
Formula: $P_2 = P_1 \times (d_1/d_2)^n$

$= 64 \times (1/2)^4$
$= 64 \times 1/16$
= 4 W
Ans: 4 W

8.
Given:
New radius = 2 × Original radius
Path-loss exponent = 2
Formula: Power factor = $(R_2/R_1)^n$

$= (2/1)^2$
= 4
Ans: 4

9.
Given:
No. of cells = 4
Channels per cell = 15

Formula: Total channels = No. of cells × Channels per cell

= 4 × 15
= 60
Ans: 60

10.
Given:
Velocity = 100 km/h
Carrier frequency = 900 MHz
Speed of light = 3×10^8 m/s
Velocity = 100 × 1000 / 3600
= 27.78 m/s
Formula: Doppler shift = (v × f)/c
= $(27.78 \times 900 \times 10^6)/(3 \times 10^8)$
= 83.33 Hz
Ans: 83.33 Hz.

11.
Given:
Frame duration = 5 ms
No. of slots = 8
Formula: Slot duration = Frame duration / No. of slots

= 5 / 8
= 0.625 ms
Ans: 0.625 ms.

12.
Given:
Total bandwidth = 24 MHz = 24000 kHz
Carrier spacing = 200 kHz
Formula: No. of carriers = Total bandwidth / Carrier spacing

= 24000 / 200
= 120
Ans: 120

13.
Given:
Transmitted power = 8 W = 8000 mW
Path loss = 33 dB
Formula: Power in dBm = 10 log10(P in mW)
= 10 log10(8000)
= 39.03 dBm
Received power = Transmitted power - Path loss
= 39.03 - 33
= 6.03 dBm
Ans: 6.03 dBm.

14.
Given:
Frequency = 1900 MHz
Distance = 2.5 km
Formula: FSPL = 32.44 + 20 log10(f) + 20 log10(d)
= 32.44 + 20 log10(1900) + 20 log10(2.5)
= 32.44 + 65.58 + 7.96
= 105.98 dB
Ans: (105.98) Round off value 106 dB.

15.
Given:
No. of cells = 6
Calls per cell = 250
Formula: Total capacity = No. of cells × Calls per cell
= 6 × 250
= 1500
Ans: 1500

16.
Given:
Call arrival rate = 4000 calls/S
Average call duration = 3 min = 180 Ss
Formula: Active calls = Arrival rate × Avg duration
= 4000 × 180
= 720000
Ans: 720000

17.
Given:
No. of channels = 150
Bandwidth per channel = 25 kHz
Formula: Total bandwidth = No. of channels × Bandwidth per channel
= 150 × 25
= 3750 kHz
= 3.75 MHz
Ans: 3.75 MHz.

18.
Given:
No. of frequency bands = 30
Guard band = 40 kHz
No. of gaps = 30 - 1 = 29

Formula: Total guard bandwidth = No. of gaps × Guard bandwidth
= 29 × 40
= 1160 kHz
= 1.16 MHz
Ans: 1.16 MHz.

19.
Given:
Speed = 90 km/h
Cell radius = 1.5 km
Cell diameter = 2 × Radius
= 2 × 1.5
= 3 km
Formula: No. of handoffs = Distance travelled / Cell diameter
= 90 / 3
= 30
Ans: 30

20.
Given:
Regional centres = 8
Base stations per centre = 40
Users per base station = 250
Formula: Total capacity = Centres × Base stations × Users

= 8 × 40 × 250
= 80000
Ans: 80000

21.
Given:
Total area = 400 km²
Cell radius = 400 m = 0.4 km
Approx cell area = R²

= 0.4 × 0.4
= 0.16 km²
No. of cells = Total area / Cell area
= 400 / 0.16
= 2500
Ans: 2500

22.
Given:
Total bandwidth = 6 MHz = 6000 kHz
Channel bandwidth = 30 kHz
Formula: No. of channels = Total bandwidth / Channel bandwidth

= 6000 / 30
= 200
Ans: 200

23.
Given:
No. of subscribers = 1200
Calls per subscriber per hour = 4
Average call duration = 2.5 min

Formula: Traffic = No. of subscribers × Calls per hour × Call duration in hours
= 1200 × 4 × (2.5/60)
= 1200 × 4 × 0.04167
= 200 Erlangs
Ans: 200

24.
Given:
Total bandwidth = 40 MHz = 40000 kHz
Bandwidth per user = 400 kHz
Formula: No. of users = Total bandwidth / Bandwidth per user
= 40000 / 400
= 100
Ans: 100

25.
Given:
No. of base stations = 1200
Capacity per base station = 600 users
Formula: Total capacity = No. of base stations × Capacity per station
= 1200 × 600
= 720000
Ans: 720000